Gwendal Fouché

Doctor in Virtual Reality and Data Visualisation

Personal Information

gwendal.fouche.pro@gmail.com

127 rue d'Antrain, 35700 Rennes

gwendalfouche.github.io

French *(native)*, English C2, Spanish B1

Technical skills

Unity, C# 8.0, .Net standard 2.0

C++ 17, R, Python

Shaders : HLSL, CG, GPU computation

Deep Learning : CNN, RNN, GAN, Auto-Encoders

OpenXR, Git

3D Modelling: Blender (certification with *OneLearn*)

Associative experience

2016 & 2017 : Volunteering at Yaouank festival

2017 : President of the *Club Kebab Supélec*

2017 : Treasurer of the association *Fest-Noz Supélec*

Professional experience

Post-doctoral researcher, AIMAC team, CentraleSupélec — de 2023 to 2025

Teaching:

- Data Visualisation, Mixed Reality (Master's – Year 2)
- Serious Game, Computer Vision (Master's – Year 1)

Research:

- *Immersive Analytics*: 3D interaction for application control and data visualisation
- Immersion as a support for *Affective Computing*
- Statistical analysis of clinical study results

Master's Final-year internship, followed by a PhD, Hybrid Research team, Inria Rennes — 2019 to 2022

Thesis title: *Immersive Interaction and Visualization of Temporal 3D Data*

- Main research in 3D interaction and 3D temporal data visualisation
- Collaboration with researchers in biology

Method prototyping:

- VR application development with Unity Engine
- Shader development for volumetric 3D data display
- GPU computation in real time for large 3D temporal datasets

Teaching assistant — 2020 to 2022

- INSA Rennes: Introduction to algorithms (Undergraduate – Year 1)
- CentraleSupélec: Virtual and Augmented Reality (Master's – Year 1)
- Ecole Polytechnique: Virtual Reality (Master's – Year 2)

Internship in FAST research team, CentraleSupélec — 2018

Project: *Acquisition of eye movements on FOVE virtual reality equipment*

Academic training

CentraleSupélec Engineering School, Supélec Curriculum — 2016 to 2019

- Electronics, electrical engineering
- Signal and Image analysis
- Major Secure Information Systems

Student projects :

- 3D graph visualization of security events via virtual reality
- Game therapy for tinnitus in virtual reality, with the startup *Immersive Therapy*

Semester at KTH, Stockholm — 2018

Master of Electrical Engineering

Classe Préparatoire aux Grandes Ecoles (MPSI, MP) — 2014 to 2016

Lycée Chateaubriand, Rennes

Research activities

Main publications

PhD Thesis : Immersive and Interactive Visualisation of 3D Temporal Data — 2023

PhD executed and financed through Inria Challenge *NAVISCOPE : image-guided NAvigation and VIsualization of large data sets in live cell imaging and microSCOPy*
Direction: Ferran Argelaguet, Emmanuel Faure, Charles Kervrann
Examiner: Christophe Hurter, Tobias Isenberg

*Immersive and Interactive Visualization of 3D Spatio-Temporal Data using a Space Time Hypercube — 2023

Gwendal Fouché, Ferran Argelaguet, Emmanuel Faure, Charles Kervrann. *Frontiers in Bioinformatics, 2023, vol. 3.*

*Timeline Design Space for Immersive Exploration of Time-Varying Spatial 3D Data — 2022

Gwendal Fouché, Ferran Argelaguet, Emmanuel Faure, Charles Kervrann. *In Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology (pp. 1-11), 2022.*

Complete list of publications and details at gwendalfouche.github.io/research.html

Participations in research community

Poster presentation at conference **NEUBIAS 2020, Bordeaux**

Research article presentation at conference **ACM VRST 2022, Tsukuba, Japon**

Reviews of research articles for conferences **IEEEVR 2022, Christchurch, Nouvelle Zélande** (associée au journal IEEE TVCG), and **ACM SUI 2023, Sydney, Australie**

Demonstration of the 3D temporal data visualisation VR application from my research works, for biology researchers from – or partners of – Curie Institute, Paris, July 2022

Participation to the *Fête de la science* at Curie Institute, Paris, Virtual Reality demonstrations for the general public and Curie Institute personnel, October 2022